

Soil Aggregation v/s AMF and Cover Crops

Anurag Banerjee

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The goal of this analysis is to see how soil aggregation metrics like water sieving... are affected by cover crops and AMF in both Fusarium and non Fusarium infected crops. For this we will use a split plot design to analyze the data. We have multiple aggregation metrics but we will focus on WIN... for now and use that as our response. The plot has been divided into 4 whole plots by cover crop and Fusarium infected based on the following levels: 1. Cover Crop and Fusarium (C-F) 2. Cover Crop and Not infected by Fusarium (C-nF) 3. No Cover Crop and Fusarium (nC-F) 4. No Cover Crop and Not infected by Fusarium (nC-nF)

(These are the whole plots)

I think each whole plot should be modeled as a random effect and not just a fixed effect.

Each whole plot is divided into 3 split plots based on the 3 levels of AMF.

The statistical model is:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_l + (\alpha\beta)_{ij} + u_{ik} + \epsilon_{ijk}$$

Y_{ijk} = kth replicate of the response measured in the jth split plot in the ith whole plot

α_i = fixed effect of the ith whole plot

β_j = fixed effect of the jth split plot

γ_l = nuisance factor for year

$(\alpha\beta)_{ij}$ = interaction effect of ith whole plot and jth split plot (this may not be required)

u_{ik} = random effect of the subplot k nested within ith whole plot, this follows a $N(0, \sigma_u^2)$ distribution

ϵ_{ijk} = error due to kth replicate of the response (kth subplot) measured in the jth split plot in the ith whole plot, this follows a $N(0, \sigma^2)$ distribution

$i = \{1, 2, 3, 4\}$; $j = \{1, 2, 3\}$; $k = \{1, 2, 3, 4, 5, 6, 7, 8\}$

k goes up to 4 for a single year, but since we have data for 2024 and 2025, we are assuming they are just repeated measurements hence we have 8 replications.

Checking for balanced design

```
##
##          AMF F W
##    c-f      8 8 8
##    c-nf     8 8 8
##    nc-f     8 8 8
##    nc-nf    8 8 8
```

Balanced design verified

Changed the structure to make it easier to write contrasts later on: 1: "c-f", 2: "c-nf", 3: "nc-nf", 4: "nc-f"

1: "W", 2: "AMF", 3: "F"

```
df_model <- df[complete.cases(df[, -4]),]
model <- lmer(WSA ~ Whole_Plot * Treatment + Year + (1|Plot_No), data = df_model)
summary(model)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: WSA ~ Whole_Plot * Treatment + Year + (1 | Plot_No)
## Data: df_model
##
## REML criterion at convergence: -57.5
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.2879 -0.4618 -0.0314  0.4937  2.9490
##
## Random effects:
## Groups Name Variance Std.Dev.
## Plot_No (Intercept) 0.003301 0.05745
## Residual 0.017798 0.13341
## Number of obs: 94, groups: Plot_No, 48
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    0.7466821  0.0569253 39.6759634  13.117 5.19e-16
## Whole_Plotc-nf    0.0004342  0.0781019 35.2954745   0.006 0.99560
## Whole_Plotnc-nf  -0.0918879  0.0781019 35.2954745  -1.177 0.24726
## Whole_Plotnc-f   -0.1213525  0.0781019 35.2954745  -1.554 0.12916
## TreatmentAMF      0.0686149  0.0781019 35.2954745   0.879 0.38560
## TreatmentF        0.0455205  0.0781019 35.2954745   0.583 0.56371
## Year2025         -0.0936754  0.0276065 46.6317862  -3.393 0.00142
## Whole_Plotc-nf:TreatmentAMF  0.0082388  0.1120907 36.6360291   0.074 0.94181
## Whole_Plotnc-nf:TreatmentAMF -0.0643053  0.1104528 35.2954745  -0.582 0.56413
## Whole_Plotnc-f:TreatmentAMF -0.1431090  0.1104528 35.2954745  -1.296 0.20350
## Whole_Plotc-nf:TreatmentF    0.0225509  0.1104528 35.2954745   0.204 0.83940
## Whole_Plotnc-nf:TreatmentF  -0.0915605  0.1120907 36.6360291  -0.817 0.41930
## Whole_Plotnc-f:TreatmentF   -0.0925529  0.1104528 35.2954745  -0.838 0.40770
##
## (Intercept) ***
## Whole_Plotc-nf
## Whole_Plotnc-nf
## Whole_Plotnc-f
## TreatmentAMF
## TreatmentF
## Year2025 **
## Whole_Plotc-nf:TreatmentAMF
## Whole_Plotnc-nf:TreatmentAMF
## Whole_Plotnc-f:TreatmentAMF
## Whole_Plotc-nf:TreatmentF
## Whole_Plotnc-nf:TreatmentF
## Whole_Plotnc-f:TreatmentF
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Cell means: $\mu = (\mu_{11}, \mu_{12}, \mu_{13}, \mu_{14}, \mu_{21}, \mu_{22}, \mu_{23}, \mu_{24}, \mu_{31}, \mu_{32}, \mu_{33}, \mu_{34}, \mu_{41}, \mu_{42}, \mu_{43}, \mu_{44})^T$

If C is a vector of contrasts, and μ is the vector of cell means, we write contrasts in the form of $c\mu$, s.t. this linear function of μ is estimable.

In Fusarium and non-Fusarium infected crops we will compare the joint effects of AMF and Cover crop separately, i.e. one set of contrasts for infected and one set for non-infected:

(We are seeing if there's a difference since we wanna see the change in joint or interaction effects of AMF and cover crops)

1. Test the difference in AMF - placebo (W) for cover crop and Fusarium and AMF - placebo (W) for no cover crop in Fusarium, i.e.

$$H_0 : \frac{\mu_{12} - \mu_{11}}{2} - \frac{\mu_{42} - \mu_{41}}{2} = 0$$

This contrast is AMF vs W in Fusarium, averaged over cover crop

$$H_a : \frac{\mu_{12} - \mu_{11}}{2} - \frac{\mu_{42} - \mu_{41}}{2} \neq 0$$

```
emm <- emmeans(model, ~ Whole_Plot * Treatment)
joint_contrast <- contrast(
  emm,
  method = list(
    "Does cover crop modify the AMF vs W effect under Fusarium?" =
      c(-1, 1, 0,
        0, 0, 0,
        0, 0, 0,
        1, -1, 0) / 2
  )
)

summary(joint_contrast, infer = TRUE)
```

```
## contrast estimate SE
## Does cover crop modify the AMF vs W effect under Fusarium? 0.103 0.0561
## df lower.CL upper.CL t.ratio p.value
## 36.2 -0.0103 0.217 1.844 0.0733
##
## Results are averaged over the levels of: Year
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

2. AMF v/s F w.r.t cover crop in Fusarium infected:

$$H_0 : \frac{\mu_{12} - \mu_{13}}{2} - \frac{\mu_{42} - \mu_{43}}{2} = 0$$

This contrast is AMF vs W in Fusarium, averaged over cover crop

$$H_a : \frac{\mu_{12} - \mu_{13}}{2} - \frac{\mu_{42} - \mu_{43}}{2} \neq 0$$

```
emm <- emmeans(model, ~ Whole_Plot * Treatment)
joint_contrast <- contrast(
  emm,
  method = list(
    "Does cover crop modify the AMF vs F effect under Fusarium?" =
      c(0, 1, -1,
        0, 0, 0,
        0, 0, 0,
        0, -1, 1) / 2
  )
)
```

```
summary(joint_contrast, infer = TRUE)
```

```
## contrast estimate SE
## Does cover crop modify the AMF vs F effect under Fusarium? 0.0309 0.0561
## df lower.CL upper.CL t.ratio p.value
## 36.2 -0.0828 0.145 0.552 0.5847
##
## Results are averaged over the levels of: Year
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

3. AMF v/s average of F and W in Fusarium infected crops

$$H_0 : \frac{\mu_{12} - \frac{\mu_{11} + \mu_{13}}{2}}{2} - \frac{\mu_{42} - \frac{\mu_{41} + \mu_{43}}{2}}{2} = 0$$

$$H_a : \frac{\mu_{12} - \frac{\mu_{11} + \mu_{13}}{2}}{2} - \frac{\mu_{42} - \frac{\mu_{41} + \mu_{43}}{2}}{2} \neq 0$$

This contrast is AMF - average of F and W

```
emm <- emmeans(model, ~ Whole_Plot * Treatment)
joint_contrast_amf_f_w <- contrast(
  emm,
  method = list(
    "Does cover crop modify the AMF vs average(F, W) effect under Fusarium?" =
      c(-1/2, 1, -1/2,
        0, 0, 0,
        0, 0, 0,
        1/2, -1, 1/2)
  )
)
```

```
summary(joint_contrast_amf_f_w, infer = TRUE)
```

```
## contrast
## Does cover crop modify the AMF vs average(F, W) effect under Fusarium?
## estimate SE df lower.CL upper.CL t.ratio p.value
## 0.134 0.0976 36.6 -0.0635 0.332 1.376 0.1771
##
## Results are averaged over the levels of: Year
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

4. Test the difference in AMF - placebo (W) for cover crop and Fusarium and AMF - placebo (W) for no cover crop in non-Fusarium, i.e.

$$H_0 : \frac{\mu_{22} - \mu_{21}}{2} - \frac{\mu_{32} - \mu_{31}}{2} = 0$$

This contrast is AMF vs W in Fusarium, averaged over cover crop

$$H_a : \frac{\mu_{22} - \mu_{21}}{2} - \frac{\mu_{32} - \mu_{31}}{2} \neq 0$$

```
emm <- emmeans(model, ~ Whole_Plot * Treatment)
joint_contrast <- contrast(
  emm,
  method = list(
    "Does cover crop modify the AMF vs W effect under non-Fusarium infected?" =
      c(0, 0, 0,

```

```

      -1, 1, 0,
      1, -1, 0,
      0, 0, 0) / 2
    )
  )

summary(joint_contrast, infer = TRUE)

## contrast
## Does cover crop modify the AMF vs W effect under non-Fusarium infected?
## estimate      SE    df lower.CL upper.CL t.ratio p.value
##    0.149 0.0552 34.8    0.037    0.261    2.700 0.0106
##
## Results are averaged over the levels of: Year
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95

```

5. AMF v/s F w.r.t cover crop in non-Fusarium infected:

$$H_0 : \frac{\mu_{22} - \mu_{23}}{2} - \frac{\mu_{32} - \mu_{33}}{2} = 0$$

This contrast is AMF vs W in non-Fusarium, averaged over cover crop

$$H_a : \frac{\mu_{22} - \mu_{23}}{2} - \frac{\mu_{32} - \mu_{33}}{2} \neq 0$$

```

emm <- emmeans(model, ~ Whole_Plot * Treatment)
joint_contrast <- contrast(
  emm,
  method = list(
    "Does cover crop modify the AMF vs F effect under non-Fusarium?" =
      c(0, 0, 0,
        0, 1, -1,
        0, -1, 1,
        0, 0, 0) / 2
  )
)

summary(joint_contrast, infer = TRUE)

## contrast
## Does cover crop modify the AMF vs F effect under non-Fusarium?      estimate      SE
## df lower.CL upper.CL t.ratio p.value
## 36.2 0.00262    0.23    2.075 0.0452
##
## Results are averaged over the levels of: Year
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95

```

6. AMF v/s average of F and W in non-Fusarium infected crops:

$$H_0 : \frac{\mu_{22} - \frac{\mu_{21} + \mu_{23}}{2}}{2} - \frac{\mu_{32} - \frac{\mu_{31} + \mu_{33}}{2}}{2} = 0$$

$$H_a : \frac{\mu_{22} - \frac{\mu_{21} + \mu_{23}}{2}}{2} - \frac{\mu_{32} - \frac{\mu_{31} + \mu_{33}}{2}}{2} \neq 0$$

This contrast is AMF - average of F and W

```

emm <- emmeans(model, ~ Whole_Plot * Treatment)
joint_contrast_amf_f_w <- contrast(

```

```

emm,
method = list(
  "Does cover crop modify the AMF vs average(F, W) effect under Fusarium?" =
    c(0, 0, 0,
      -1/2, 1, -1/2,
      1/2, -1, 1/2,
      0, 0, 0)
)
)

summary(joint_contrast_amf_f_w, infer = TRUE)

```

```

## contrast
## Does cover crop modify the AMF vs average(F, W) effect under Fusarium?
## estimate      SE    df lower.CL upper.CL t.ratio p.value
##      0.265 0.0962 35.3   0.0703    0.461   2.761  0.0091
##
## Results are averaged over the levels of: Year
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95

```